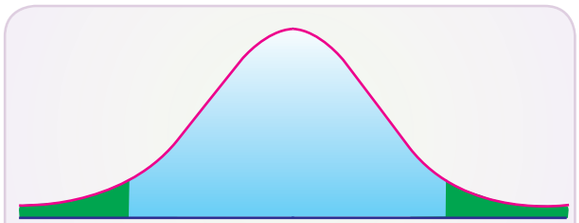
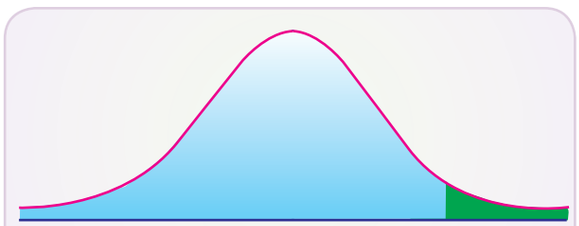
of the tails of the sampling distribution, then it is known as one-tailed test. Here $H_1 : \mu > \mu_0$ and $H_1 : \mu < \mu_0$ are known as one tailed alternative.



Two tailed diagram

Fig.8.1

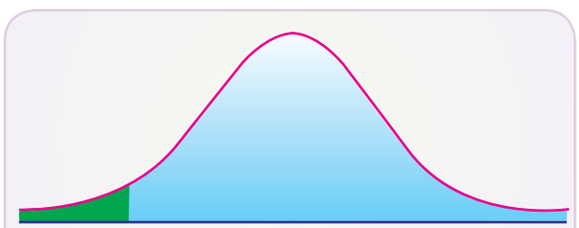
Right tailed test: $H_1 : \mu > \mu_0$ is said to be right tailed test where the rejection region or critical region lies entirely on the right tail of the normal curve.



Right tailed diagram

Fig.8.2

Left tailed test: $H_1 : \mu < \mu_0$ is said to be left tailed test where the critical region lies entirely on the left tail of the normal curve. (diagram)



Left tailed test

Fig.8.3

Types of Errors in Hypothesis testing

There is every chance that a decision regarding a null hypothesis may be correct or may not be correct. There are two types of errors. They are

Type I error: The error of rejecting H_0 when it is true.

Type II error: The error of accepting when H_0 it is false.

Critical region or Rejection region

A region corresponding to a test statistic in the sample space which tends to rejection of H_0 is called critical region or region of rejection.

Note

The region complementary to the critical region is called the region of acceptance.

Level of significance

The probability of type I error is known as level of significance and it is denoted by α. The level of significance is usually employed in testing of hypothesis are 5% and 1%. The level of significance is always fixed in advance before collecting the sample information.

Critical values or significant values

The value of test statistic which separates the critical (or rejection) region and the acceptance region is called the critical value or significant value. It depend upon

(i) The level of significance

(ii) The alternative hypothesis whether it is two-tailed or single tailed.

For large samples, the standardized variable corresponding to the statistic viz.,

$$Z = \frac{t - E(t)}{\sqrt{Var(t)}} = \frac{t - E(t)}{S.E.(t)} \sim N(0,1) \text{ as } n \to \infty \quad \ldots (1)$$

The value of Z given by (1) under the null hypothesis is known as test statistic. The critical values of Z, commonly used at the level of significance for both two tailed and single tailed tests are given in the normal probability table (Refer the normal probably Table).

Since for large n, almost all the distributions namely, Binomial, Poisson, etc., can be approximated very closely by a normal probability curve, we use the normal test of significance for large samples.

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